

Linear eq.

Non linear eq.

$$a + b = 10$$

$$a^2 + b^2 = 16$$

$$2a + 3b = 15$$

$$8i(a) + b^2 = 15$$

$$3.4a + 48.9b + 22 = 122$$

$$ab^2 + \frac{b}{a} - \frac{3}{b} = \log a$$

Only Numbers allowed

$$\begin{cases} a + b = 10 \\ 2a + 2b = 20 \end{cases}$$

$$2(a + b) = 2(10)$$

$\therefore$ Not solvable

2a

$$(a + b) \cdot 2 = 2a + 2b$$

both equations are same!

Not enough information

Redundant!

Infinitely many solutions

a	b
0	2
5	5
2	8
0	10
10	0

$$\begin{cases} a + b = 10 \\ 2a + 2b = 24 \end{cases}$$

Contradictory

incorrect info.

$$\begin{cases} a + b = 10 \\ a + 2b = 12 \end{cases}$$

Complete
Non-singular

$$\begin{cases} a + b = 10 \\ 2a + 2b = 20 \end{cases}$$

Redundant
Singular

$$\begin{cases} a + b = 10 \\ 2a + 2b = 24 \end{cases}$$

Contradictory
Singular

A system of sentences, system of equations is same.

e.g)

- ① apple & banana cost \$10
 ② apple & 2 banana cost \$12

$$a + b = 10, \quad a + 2b = 12$$

$$\begin{array}{r} a + 2b = 12 \\ - a + b = 10 \\ \hline 0 + b = 2 \end{array} \quad \therefore \boxed{b=2} \quad \boxed{a=8}$$

$$\begin{array}{r} 0 + b = 2 \\ \rightarrow a + 2b = 12 \\ a + 2(2) = 12 \rightarrow \boxed{a=8} \end{array}$$

$$\boxed{a + b + c = 10}, \quad \boxed{a + 2b + c = 15},$$

$$\boxed{a + b + 2c = 12}$$

$$\begin{array}{r} a + 2b + c = 15 \\ - a + b + c = 10 \\ \hline b = 5 \end{array}$$

$$\begin{array}{r} a + b + c = 10 \\ a + 5 + c = 10 \\ \hline a + c = 5 \end{array}$$

$$\boxed{b=5}$$

$$\boxed{a+c=5}$$

$$\begin{array}{r} a + 2b + c = 15 \\ 2 \times a + b + 2c = 12 \\ \hline -a + 3c = 3 \end{array}$$

$$\begin{array}{r} -a + 3c = 3 \\ + a + 2 = 3 \\ \hline c = 1 \end{array}$$

$$\begin{array}{r} a + b + 2c = 12 \\ a + 5 + c = 10 \\ \hline c = 2 \end{array}$$

$$\boxed{c=2}$$

$$a + c = 5 \rightarrow \boxed{a=3}$$

$$a + 2 = 5$$

Designing another system of Sentences

① Among dog, bird, cat → one is red

② Among dog, cat → one is orange

③ Dog is black

So → Dog = black

dog = black, cat = orange

dog = b, cat = or, bird = red

So this system gives complete info

Complete sentences

∴ No redundancy
2
No contradictions

Non-singular

Means every sentence provides a unique information and no two sentences contradict

System of sentences

Assuming I have a cat & dog & a bird
& have one colour ^{only} each

System 1

Dog is black
cat is orange
Bird is red

↓
Complete Sentence
(has all info)
& each sentence
is definable)

↓
Non-Singular

Sys 2

Dog black
Cat orange
Dog black

↓
~~Info~~
Redundant
(contains same
info)

↓
Singular

Sys 3

Dog black
Dog black
Dog black

↓
Redundant
(same info)

↓
Singular

Sys 4

Dog black
Dog white
Bird red

↓
Contradictory
(2 pieces of
info contradict)

↓
Singular

This system
is less redundant
than this

but how much?
for the we calculate
the (rank of the matrix)
later.